

Problem Set 1 – Temporal Profile Analyzer

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Source Code

```
1  """
2  Program: Temporal Profile Analyzer
3  Purpose: Computes an AI Era Readiness Score from a name and age.
4  Author: Kshitiz
5  """
6
7  import datetime
8
9  SINGULARITY_YEAR = 2045
10
11  # --- Name Analysis ---
12  user_full_name = input("Enter your full name: ").strip()
13
14  if not user_full_name:
15      print("Input Error: Name cannot be empty or whitespace-only.")
16  else:
17      name_length = len(user_full_name)
18      formatted_name = user_full_name.title()
19      print(f"Identifier Byte-Count: {name_length}")
20      print(f"Formatted Name: {formatted_name}")
21
22  # --- Temporal Projection ---
23  age_input = input("Enter your current age: ").strip()
24
25  if not age_input.isdigit():
26      print("Input Error: Age must be a valid non-negative whole number.")
27  else:
28      current_age = int(age_input)
29      current_year = datetime.date.today().year
30      age_in_2045 = current_age + (SINGULARITY_YEAR - current_year)
31      print(f"Current Year: {current_year}")
32      print(f"Projected Age in {SINGULARITY_YEAR}: {age_in_2045}")
33
34  # --- AI Readiness Formula ---
35  score = ((name_length * 10) + age_in_2045) / 2
36  print(f"AI Era Readiness Score: {score:.2f}")
37
38  # Bonus: repeat name by age's leading digit.
39  # age // 10 gives 0 for single-digit ages, so just use age then.
40  if current_age >= 10:
41      repeat_count = current_age // 10
42  else:
43      repeat_count = current_age
44
45  repeated_name = formatted_name * repeat_count
46  print(f"Bonus - Repeated Name ({repeat_count}x): {repeated_name}")
```

Console Log

```
1  --- Run 1: Perfect run (matches worked example) ---
2  Enter your full name: Scooby-Dooo
```

```

3 Identifier Byte-Count: 12
4 Formatted Name: Scooby-Dooo
5 Enter your current age: 20
6 Current Year: 2026
7 Projected Age in 2045: 39
8 AI Era Readiness Score: 79.50
9 Bonus - Repeated Name (2x): Scooby-DoooScooby-Dooo
10
11 --- Run 2: Failed run (non-numeric age) ---
12 Enter your full name: Kshitiz Reddy
13 Identifier Byte-Count: 13
14 Formatted Name: Kshitiz Reddy
15 Enter your current age: twenty
16 Input Error: Age must be a valid non-negative whole number.

```

Submission Notes

Name is checked after `.strip()`, so a blank/whitespace-only name is caught by `not user_full_name` before the length calc runs. Age uses `str.isdigit()` before `int()`, so bad input like "twenty" never reaches the conversion. Current year comes from `datetime.date.today().year` so the 2045 projection stays correct regardless of when this runs.

For the bonus, repeat count is normally `age // 10`. That gives 0 for single-digit ages, so the script falls back to using the age itself when `current_age < 10`.